LLM-Assisted Explanation of Machine Learning Model Outputs: The Effect of XAI Method and Handover Format on Explanation Quality

Term paper

Anton Stahl

Technical University of Dresden
Chair of Business Information Systems, esp. Intelligent Systems and Services
Email: anton_paul.stahl@mailbox.tu-dresden.de
Supervisor: Prof. Dr. Patrick Zschech

Abstract. Explainable AI (XAI) outputs such as SHapley Additive exPlanations (SHAP) beeswarms, shape functions and dependence plots are hard to read for non-experts, and large language models (LLMs) can translate them into prose. How the artifact is handed to the LLM (as numbers, as an image, or through tool calls) has not been studied systematically. This paper compares three such handovers and a deterministic template on two models trained on UCI Bike Sharing data: XGBoost explained with TreeSHAP and an inherently interpretable Explainable Boosting Machine (EBM). It scores 72 per-feature and ten whole-model descriptions against a mechanically derived ground truth, using a deterministic rubric and two LLM judges from different vendors. The two judges agree only weakly on faithfulness (Krippendorff's α 0.30–0.48). Four results stand. First, on the per-feature task, repeated generation reorders the three LLM handovers. Resampling produces all six possible orderings, so a single draw cannot show which format leads. Second, on the whole-model task and on the rubric, the LLM reads a beeswarm image about as well as the same content given as numbers, but a beeswarm alone loses the shape detail that nine per-feature plots carry. Third, on the same task and instrument, letting the LLM pull artifacts through tools does not beat pushing them as numbers. Fourth, on the per-feature task the template comparison depends on the judge. The Anthropic judge finds no difference, partly because it penalizes a phrasing flaw of the template. The OpenAI judge favors the template, but only one contrast survives correction and it rests on a single verdict. In practice, the image handover needs 1 640 instead of 164 360 input tokens per XGBoost feature without a detectable loss in faithfulness, and a single LLM judge is not a sufficient faithfulness check.

Keywords: Explainable AI, SHAP, EBM, LLM-as-a-Judge, Design Science Research

1 Introduction

1.1 Motivation

A machine learning model that forecasts hourly bike demand does not make decisions, but it produces numbers that people who are not data scientists have to read and act on. When those readers ask why the numbers are what they are, the standard explainable AI (XAI) libraries return a beeswarm plot, a shape function or a stack of dependence scatters, each dense with information and hard to read at a glance. Kaur et al. (2020) observed that even data scientists working with InterpretML and SHAP (SHapley Additive exPlanations) over-trusted and misused such visualizations: they rarely described them accurately, took them at face value and used them to rationalize suspicious observations. If trained users misread these plots, the operational reader has little chance. The EU AI Act (European Union 2024, Art. 86) sharpens the problem without operationalizing it. It gives a person affected by a decision based on a high-risk AI system the right to "clear and meaningful explanations" of the system's role, but leaves the standard for "clear and meaningful" unspecified.

One route out of this is a translation layer. A large language model (LLM) can turn a plot or a table of contributions into a short piece of prose that a non-expert can read. Martens et al. (2025) report that over 90% of a general-audience sample find such story-based explanations convincing. Bordt et al. (2024) and Lengerich et al. (2023) show that LLMs can describe the shape functions of a glass-box model by feature and reason across them. Convincing is not the same as faithful: a fluent explanation is not automatically true to the model, and it motivates measuring the translation layer rather than merely trusting it.

1.2 Problem Statement

Three gaps in the existing literature define the space this paper works in. First, the work that evaluates LLM-generated explanations rigorously explains single predictions (Martens et al. 2025, Ichmoukhamedov et al. 2024), while the work that describes a model as a whole (Bordt et al. 2024, Lengerich et al. 2023) does not evaluate the descriptions against a reference, although model behavior as a whole is the operationally relevant unit for planning and audit. Second, a recent factorial study of LLM-generated XAI explanations (Lukassen, Herrmann, Weisser, Saefken & Kneib 2026) varies the forecasting model, the XAI method, the LLM and the prompting strategy, but holds fixed the channel through which the XAI output reaches the LLM. The handover channel at constant model and constant prompt is unstudied. Third, multimodal LLMs still fall far short of humans on realistic, complex charts (Wang et al. 2024), but whether this holds for the beeswarm plot has not yet been tested. A related user study by Kruschel et al. (2026) reports preliminary results ($N = 42$) in which participants with conversational-XAI assistance and with Q&A-based assistance both outperformed the model, with no measurable difference between the two forms. This leaves open whether the form of assistance or the explanation both groups shared limits performance. This paper studies the explanation side.

1.3 Research Questions

Five research questions structure the work. Their order is narrative, not evidential:

RQ1 (Representation). How much of the information carried by nine per-feature plots does a single beeswarm carry, and does the LLM read the beeswarm image as well as the equivalent numeric summary?

RQ2 (Baseline). What does the LLM add over a deterministic template that writes out the ground-truth record in fixed phrases?

RQ3 (Handover format). How does the form of handover (numeric, visual, agentic) affect faithfulness?

RQ4 (Mechanism). Does pulling the artifacts through tools differ from having them pushed, when pull can reach at least the information push delivers?

RQ5 (XAI method). How do the shape functions of an Explainable Boosting Machine (EBM) and post-hoc SHAP compare as an explanation basis, and does the generation pipeline interact with that choice?

RQ1 and RQ4 are answered on the whole-model track (one description of all nine features), RQ2 and RQ5 on the per-feature track (one description per feature), and RQ3 on both (Section 3.2). RQ5 compares the two bases as used in practice, since XGBoost needs a post-hoc method such as TreeSHAP to become interpretable at all. The results must still be read as descriptive rather than causal: the XAI method is confounded with the prediction model (XGBoost pairs with TreeSHAP, EBM pairs with its shape functions).

1.4 Contribution

This paper makes three contributions. First, an evaluation setup for LLM-generated global explanations. Four generation pipelines (a deterministic template and three LLM handovers: JSON, vision, tool use) run on a glass-box arm and a post-hoc arm with a byte-identical prompt core, and every description is scored against a mechanically derived ground truth by a deterministic rubric and a two-vendor judge. Second, a variance check showing that an ordering of LLM handovers read off a single draw cannot be interpreted. Third, empirical results on representation, mechanism and the template baseline, each stated with the track and the instrument that carry it. In brief, a single draw cannot rank the three LLM handovers, and whether the template matches the LLMs depends on the judge. On the rubric, the beeswarm image is read about as well as the equivalent numbers, and pulling artifacts through tools does not beat pushing them. Alongside these, the paper constructs a beeswarm from an EBM's exact additive contributions, so that both arms can be handed over with the same artifact (Section 4.4).

1.5 Structure

Section 2 reviews the literature, Section 3 sets out the research design and Section 4 the implementation. Section 5 reports the results, Section 6 discusses them and Section 7 concludes.

2 Theoretical Background

2.1 Interpretability: Post-Hoc vs. Glass-Box

The field distinguishes two ways to make a model interpretable: build a model that is interpretable by construction, or attach an interpretation to an opaque one after the fact. Rudin (2019) argues, for consequential decisions, for the first: an inherently interpretable model whose behavior can be inspected directly is preferable to a black box wrapped in a post-hoc approximation. This axis is what makes XAI method a meaningful factor of comparison here.

On the post-hoc side, the unifying framework is SHAP (Lundberg & Lee 2017). SHAP assigns each feature a contribution to a specific prediction such that the contributions sum to the model's output, a property called local accuracy ($f(x) = \phi_0 + \sum_i \phi_i x'_i$). Together with missingness and consistency, it uniquely characterizes the Shapley values. Exact Shapley values are exponential in the number of features in general. TreeSHAP (Lundberg et al. 2019) exploits tree structure to compute them exactly and in polynomial time, here with the tree-path-dependent value function (the `TreeExplainer` default without background data). This makes the XGBoost attributions, and with them the ground truth on the XGBoost arm, exact and deterministic.

On the glass-box side, GA^{2}M (Lou et al. 2013) is an additive model of univariate shape functions plus pairwise interactions, implemented in the InterpretML library (Nori et al. 2019) as the Explainable Boosting Machine (EBM). It has the compact form

$$g(\mathbb{E}[y]) = \beta_0 + \sum_j f_j(x_j) + \sum_{(j,k)\in\mathcal{I}} f_{jk}(x_j, x_k), \tag{1}$$

where g is the link function, f_j is the learned shape function for feature j, and $\mathcal{I}$ indexes the interaction pairs the model chose to include. Because each f_j is a first-class object the library plots directly, an EBM needs no external explanation for its main effects. The plot is the main-effect part of the model. The pairwise interaction terms f_{jk} sit next to it and carry 31% of the term importance on this dataset (Section 4.4).

Global effects can also be read off a post-hoc model through partial dependence (Friedman 2001) or, under correlated predictors, Accumulated Local Effects (Apley & Zhu 2019). Neither device is used for the global track. The EBM's own f_j is read from `explain_global()` directly, and the XGBoost feature effect is the SHAP dependence scatter for that feature, one point per training row. Partial dependence appears only in the instance-level pre-study (Appendix E). Plot semantics throughout follow Molnar (2025).

The beeswarm, the central artifact of this paper, encodes four channels simultaneously (Lundberg et al. 2019): row order = global importance, horizontal position = the instance's SHAP value, color = the feature value, horizontal spread = the heterogeneity of the effect. This density is what makes it powerful for a data scientist and hard to read for anyone else, and it is the reason it is a good target for a translation layer.

Finally, the taxonomy of Barredo Arrieta et al. (2020) locates this paper. It lists text explanations and feature-relevance explanations as separate classes of post-hoc technique. This paper turns feature-relevance outputs into text explanations and applies them globally rather than locally.

2.2 LLMs for XAI

Two direct predecessors to this paper are TalkToEBM (Bordt et al. 2024) and the hierarchical description of glass-box graphs by Lengerich et al. (2023). The first shows that an LLM can describe an EBM's shape functions in prose that reads naturally. The second decomposes the whole-model description into per-feature reasoning steps that culminate in an anomaly summary. Both operate purely descriptively (they do not measure the output against a reference), and this is exactly the gap the evaluation instruments of the present paper fill for the per-feature and whole-model tracks. Narrative XAI in the SHAP-stories and counterfactual-stories tradition (Martens et al. 2025) is the other predecessor, and it evaluates on user perception rather than on faithfulness.

Conversational and agentic XAI is represented most concretely by TalkToModel (Slack et al. 2023), which couples an LLM to an execution component that runs XAI queries on demand. This is the reference architecture for the tool-use handover in this paper, though with a fixed system prompt and a small tool catalog rather than an open-ended dialogue. A more elaborate agentic line is He & Martens (2026), whose Narrator/Critic loop iteratively refines an LLM's XAI narrative against faithfulness and coherence critics across five LLMs and five tabular datasets. This comes from the same group as Martens et al. (2025) and is the closest published counterpart to the pull condition studied here, though in a more elaborate multi-agent variant. A parallel line of work (Singh et al. 2023) builds LLM knowledge into the interpretable model rather than the explanation of it, which is a different lever and not the route taken here.

Several information-systems papers frame why the translation layer is worth studying beyond the pure ML question. Spitzer et al. (2024) run a user study on augmentation strategies. Jeck et al. (2025) address personalization and addressee fit, which is what justifies stipulating a non-technical addressee in the prompt and using clarity as one of the judge criteria. Bokstaller et al. (2025) propose a reference architecture for XAI chatbots. Brasse et al. (2023) survey the state of XAI in information systems. Bauer et al. (2023) study how XAI changes information processing. Finally, the preliminary null of Kruschel et al. (2026) between conversational and Q&A-based assistance (Section 1.2) motivates asking whether the explanation-quality side is where the gap sits.

Prompting foundations enter in three places. The `<analysis>` scratchpad that the three LLM pipelines write before answering is prompted chain-of-thought (CoT) reasoning (Wei et al. 2022). Extended thinking is not enabled, so this scratchpad is the model's only reasoning step. The agentic loop underlying the tool-use pipeline is ReAct (Yao et al. 2023). A recent taxonomy of prompt-engineering techniques (Schulhoff et al. 2025) informs the decision to keep the system core minimal and to vary only the handover block.

2.3 Evaluating LLM-Generated Explanations

Zytek et al. (2024) rate LLM-generated XAI narratives on five metrics: soundness, fluency, completeness, context-awareness and length, the first four on a 0–2 scale. The judge adapts three of them and scores each from 1 to 5 (Section 3.5). Soundness becomes faithfulness and completeness keeps its name. Fluency is scored as clarity, the broader term the authors had replaced with fluency. Two concepts carry the measurement. First,

Maynez et al. (2020) distinguish faithfulness (fidelity to the source) from factuality (fidelity to the world). Throughout this work, faithfulness denotes fidelity to the model, not to the world. A statement can be factually correct and still unfaithful, if the model did not learn it. Second, Ji et al. (2024) frame hallucination as a failure class. Its form most relevant to interpretability is over-attribution: a description claims a plausible real-world effect that the model did not learn.

The rubric extends the instance-level metrics of Ichmoukhamedov et al. (2024) from single predictions to global descriptions. Their rank, sign and value agreement (RA / SA / VA) are accuracies over quantities extracted from a per-instance narrative, and quantities the narrative does not state drop out of the denominator. In the global version, rank agreement becomes a graded importance-rank score and sign agreement a direction score, and the rubric adds a structural criterion. Value agreement has no global counterpart, because a global description refers to no single instance, so the judge scores the closest global quantities (peak location and sign-change point). Unlike the original, the rubric scores a missing rank as 0, a rule that never fires because every description states a rank (72/72 and 90/90).

Zheng et al. (2023) study the LLM-as-a-judge paradigm systematically and catalog its known biases. G-Eval (Liu et al. 2023), which appeared a few months earlier, has the model generate chain-of-thought evaluation steps before it fills in a score. Both judges here follow this reason-then-score order. G-Eval decodes at $\mathrm{temperature} = 0$ only in its GPT-3.5 variant. Its GPT-4 variant samples 20 times at $\mathrm{temperature} = 1$ and weights the scores by their probabilities. The self-preference bias (Panickssery et al. 2024) is the reason the judge runs on two vendors. Inter-rater reliability throughout uses Krippendorff's α on the interval metric (Hayes & Krippendorff 2007).

2.4 Research Gap

The three gaps identified in Section 1.2 are visible against the literature just reviewed. First, Bordt et al. (2024) and Lengerich et al. (2023) describe globally but do not evaluate systematically against a reference. Martens et al. (2025) and Ichmoukhamedov et al. (2024) evaluate rigorously but locally. Second, Lukassen, Herrmann, Weisser, Saefken & Kneib (2026) vary XAI method, LLM and prompting simultaneously and find that the LLM choice dominates, which leaves the handover channel at constant model unstudied. Third, ChartQA (Masry et al. 2022) and Chart-to-Text (Kantharaj et al. 2022) establish chart understanding as its own task, and CharXiv (Wang et al. 2024) documents large gaps between multimodal LLMs and humans on realistic, complex charts. This line of work has not been applied to the canonical XAI artifact. The design in the next section addresses these three gaps jointly.

3 Research Design

3.1 Design Science Research Framing

The work follows the Design Science Research paradigm (Hevner et al. 2004, Peffers et al. 2007). The artifact is a pipeline that produces textual global explanations of

a trained model from that model's XAI outputs. The evaluation is the experimental comparison of the pipeline's variants against a mechanically derived reference. In the contribution taxonomy of Gregor & Hevner (2013), the work is an improvement: the problem is known (feature-attribution XAI is hard to read for non-experts), and the solution is new (an LLM translation layer whose handover channel is controlled). The six activities of Peffers et al. (2007) map onto the paper directly. The problem was identified from a supervisory pre-study. The objective was to test which handover channel best preserves faithfulness. Design and development produced four generation pipelines and a two-part evaluation instrument. Demonstration ran the pipelines on 9 features $\times$ 2 XAI models and on 10 whole-model conditions. Evaluation is Sections 5.3–5.8. Lastly, communication is this paper.

3.2 Unit of Analysis and Design Decisions

The unit of analysis is a feature $\times$ XAI model pair, not a single prediction. The local explanations were run first as a pre-study ($n = 20$ per pipeline). Three of the four pipelines saturated at the maximum faithfulness score of 5.00, leaving the metric unable to rank them: this was the empirical reason to move to the feature-level global unit rather than to defend a metric that no longer discriminated. The instance-level records are retained in Appendix E as a design-evidence artifact rather than a co-equal result.

Five further design decisions shape the study. Dataset: UCI Bike Sharing is used because its features are legible to an operational reader. Kruschel et al. (2026) use the same dataset with an EBM in their user study on conversational XAI, so this paper examines the explanation side of a comparable setup. Two tracks: the per-feature track (G2a) describes one feature per answer, the whole-model track (G2b) all nine features in one answer. RQ1 (one beeswarm against nine plots) can only be asked of a whole-model task, and RQ4 (pull against push) is answered there because the tool loop then has several features to choose between. Generator: all LLM pipelines run Claude Sonnet 4.6 (`claude-sonnet-4-6`) at $\mathrm{temperature} = 1.0$, the Anthropic API default and the setting a deployed assistant would use. The Anthropic judge (`claude-opus-4-8`) shares a model family with the generator, so a second judge from another vendor runs alongside it to hedge against self-preference (Panickssery et al. 2024). Single-channel handovers: each LLM condition receives the artifact through exactly one channel, so every condition stays identifiable. Mechanical ground truth: descriptions are scored against a ground truth derived from the trained models (Section 3.4) rather than against human ratings, so scores reproduce across runs. The judge scores faithfulness, clarity and completeness, adapted from Zytek et al. (2024) (Section 2.3), with faithfulness as the main criterion.

All five research questions were fixed before any result of the global tracks existed (RQ3 to RQ5 at the start of the project, RQ1 and RQ2 at a later supervisory meeting).

3.3 Factors and Conditions

The first factor is the XAI method. Two levels: XGBoost with TreeSHAP (post-hoc attribution on a black-box tree ensemble) and EBM (glass-box additive model, native

Table 1. The two arms of the XAI factor: what the prediction model is, what the attribution is, and how the two artifacts (per-feature curve, beeswarm) are derived on each side.

Arm	Prediction model	Attribution	Per-feature artifact	Beeswarm
Post-hoc	XGBoost (Chen & Guestrin 2016)	TreeSHAP (exact, path-dependent)	SHAP dependence scatter	SHAP standard
Glass-box	EBM (Lou et al. 2013, Nori et al. 2019)	term contributions	learned shape function f_j	constructed (§4.4)

Table 2. Information channels of the five G2b whole-model handovers. Only the beeswarm pair is matched by construction. The three `*_all` conditions differ in how the rank channel is encoded. `vision_all` has no explicit numeric rank field, but the nine images are placed in rank order and all plot titles carry per-plot importance values.

Condition	curve per feature	rank	mechanism	images
`json_all`	yes (aggregated)	explicit per feature	push	none
`vision_all`	yes (9 plots)	**implicit only** (image order, importance in plot titles)	push	9
`tooluse_all`	on request	on request	pull	on request
`json_beeswarm`	no	explicit	push	none
`vision_beeswarm`	no	visual (row order)	push	1

term contributions). Because each XAI method is paired with a matched prediction model, the factor is confounded and is read descriptively rather than causally. Section 5.3 reports it as such. Table 1 makes the mapping explicit.

The second factor is the generation pipeline: `template` (a deterministic prose baseline that reads the same ground-truth data structure), `json` (the raw curve as a JSON payload), `vision` (the shape or dependence PNG), and `tooluse` (an agentic loop that pulls what it needs from a small tool catalog). The prompt core is byte-identical across the three LLM conditions, which differ only in their handover blocks (Appendix A). Tests assert this byte-identity at prompt-assembly time, so outcome differences cannot come from the shared core.

For the per-feature track (G2a) this gives $4 \times 2 \times 9 = 72$ descriptions. For the whole-model track (G2b) the design widens to five conditions per XAI model. Each of the ten whole-model answers is split into nine records, one per feature, for scoring (90 split records). G2b has two orthogonal axes. Axis 1 (representation) contrasts nine separate per-feature plots (the `*_all` conditions) with one beeswarm (`*_beeswarm`). It is reported per ground-truth field, and no overall winner between nine plots and one beeswarm is declared, because a beeswarm cannot convey shape. Axis 2 (mechanism) contrasts push (`json_all`, `vision_all`) with pull (`tooluse_all`). Both axes are read on `fair_total`, the summary score that averages only over the ground-truth fields the representation can carry (direction, rank, structure for `*_all`, direction and rank only for `*_beeswarm`).

For the beeswarm pair the information is deliberately matched: the JSON payload carries rank, color direction and coarse spread only, matching what the beeswarm image conveys. For the three `*_all` conditions it is not. `json_all` carries a rank field per feature, `tooluse_all` can call `get_feature_importances`, and `vision_all` has no numeric rank channel and is told to judge importance from the vertical range of the plots. It nonetheless carries two implicit ordering channels: the nine images arrive in global-importance order, and each plot title prints its importance (Table 2).

3.4 Ground Truth Derivation

The 18 ground-truth references (9 features $\times$ 2 XAI models) are derived by fixed rules from each model's own effect curve: for the EBM, the shape-function grid and for XGBoost, the TreeSHAP values averaged per unique feature value. A feature's importance is its mean absolute contribution, which fixes its rank (`importance_rank`, 1–9). A single classifier (`utils.global_feature.classify_shape`), shared with the template baseline, then assigns a `form`. It first tests importance: any feature, numeric or categorical, whose importance is below `flat_importance` $= 0.025$ is labeled `near-flat`, so this label means negligible importance rather than a flat curve. Only features above that threshold are labeled by type and shape: `categorical` for categorical features, and otherwise `monotonic` or `non-monotonic`. A curve counts as monotonic if the movement against its dominant direction is small, $\min(\text{up}, \text{down})/(\text{up}+\text{down}) \leq 0.15$, where up and down are the summed rises and falls along the curve. Both thresholds are a convention rather than derived values and Section 5.2 shows that 15 of 18 form labels are stable across a grid around them. The `direction` (`increasing`, `decreasing`, `mixed`, `categorical`) and `monotonicity` fields describe the curve itself and are not overridden by the near-flat label, so a near-flat feature keeps its underlying sign. Non-monotonic features carry a `peak` (`max` for an inverted U, `min` for a U, with its x and y). Categorical features carry their three highest and lowest categories. Two further fields record the first zero crossing (`sign_change_x`) and the curve's span (`effect_range`), and an optional `note` flags a domain caveat (used on the two `weathersit` records). The judge receives all fields except `effect_range` (`utils/eval.py`, `_global_reference`). The full schema is given in Appendix B.

Because the ground truth and the template baseline share the classifier, the template hits the form field by construction. Readers should keep this in mind throughout:

> The template is not an independent competitor. It shares the classifier with the ground truth, so on the form field it matches by construction, and its direction, peak and rank come from the same derivation. It is therefore an oracle-informed baseline. Where it scores below the ceiling, the cause is its fixed phrasing, not a reading error. The LLM conditions never receive the derived class. They read the raw curve, plot or tool output and must recognize near-flat behavior themselves, so their errors on this dimension are genuine.

One domain caveat is recorded but not corrected: `weathersit` category 4 (heavy rain / thunderstorm) has only three training rows, so its high importance rank is not domain-reliable. The reference notes this rather than reranking, so that the LLMs' scores are compared against what the model actually learned, not against what it should have learned.

3.5 Measurement Instruments

Three instruments are used together. A finding is called robust when the rubric and both judges agree (the convergence rule). Where they do not, it is reported as an instrument-specific statement. The near-flat template lead of Section 5.3 meets this rule. Most other

results are instrument-specific, so each headline in the abstract, the discussion and the conclusion names the instrument it rests on.

The rubric is a deterministic scorer. It parses the `[EFFECT]` and `[IMPORTANCE]` sections with regular expressions and keyword lists and produces three sub-scores in $[0, 1]$: `direction`, `rank`, `structure`. The keyword lists were drafted with an LLM from the generator's model family and reviewed by the author (Appendix G). The rank sub-score is graded: an exact match scores 1.0, off by one 0.5, the same coarse tier (top-3 / middle / bottom-3) 0.25, and anything else or no stated rank 0.0.

The judge is a reference-grounded LLM scorer. It sees the ground-truth record and the candidate description, writes a reasoning trace and then gives a score from 1 to 5 on each of three criteria adapted from Zytek et al. (2024): faithfulness (Section 2.3), clarity (understandability for a non-technical reader) and completeness (whether the effect, importance and recommendation sections are present and substantial). It follows G-Eval's reason-then-score order (Liu et al. 2023) with an anchor-plus-deduction rule (score $= \max(1, \text{anchor} + \sum \text{deductions})$) and three in-context anchors. It runs on two vendors: OpenAI `gpt-4o` at $\text{temperature} = 0$, the setting G-Eval uses in its GPT-3.5 variant, and Anthropic `claude-opus-4-8`, which does not accept the `temperature` parameter and therefore runs at the API default. Because the judge is reference-grounded, it does not have to guess whether a curve has a peak, only whether the description matches the reference. The same grounding means that rubric and judge are not independent: their agreement is internal consistency, not criterion validity.

The third instrument is qualitative error analysis by form type: manual inspection of the records that scored lowest, cross-referenced against the ground truth. It is the third, non-numeric evidence and is used to examine the over-attribution pattern.

3.6 Statistical Procedure

Comparisons across generation pipelines use the Wilcoxon signed-rank test paired over the 18 (feature, XAI model) pairs. The pairing removes feature difficulty, the largest source of variance in the score. A parametric t-test is not used because the judge scores sit on a short 1–5 ordinal scale, the rubric sub-scores on $\{0, 0.25, 0.5, 1.0\}$, and $n = 18$ is small. The six pairwise comparisons within one judge are Holm-corrected (Holm 1979), since the uncorrected family-wise error rate would be about 26%. As a sensitivity check, Holm is also applied across both judges jointly (twelve tests). Main-text results stated against "the judge" without a vendor label refer to the Anthropic judge. Pairs with identical scores are dropped, and because judge scores are integers only 4–9 pairs remain informative on the Anthropic judge and 10–16 on the OpenAI judge (Table 16). At four informative pairs even a unanimous 0:4 split gives an exact p of 0.125, so a null on the Anthropic judge is weak evidence of no difference. p-values in the text use the normal approximation. The exact permutation values in Table 16, which also gives the paired rank-biserial correlation, change no within-judge conclusion. Effect size is Cliff's δ (Cliff 1993) at the thresholds of Romano et al. (2006) ($|\delta| < 0.147$ negligible, < 0.330 small, < 0.474 medium, otherwise large), and a positive δ means that the first-named format tends to score higher. Cross-vendor reliability uses Krippendorff's α on the interval metric (Hayes & Krippendorff 2007), and bootstrap confidence intervals use

2 000 resamples with seed $= 42$. A non-significant result is reported as "no detectable difference", never as "demonstrably equal", since no equivalence test is run.

4 Implementation

4.1 Dataset and Preprocessing

The dataset is the UCI Bike Sharing dataset (Fanaee-T & Gama 2014), an hourly record of the Capital Bikeshare rentals in Washington D.C. across 2011 and 2012, with 17 379 hourly observations. The target is `cnt`, the total rentals in that hour.

Three groups of columns are removed before training. The leakage set (`casual`, `registered`, `instant`) exposes the target. The redundancy set (`season`, `workingday`) duplicates `mnth`, `weekday` and `holiday`, and apparent temperature is dropped in favor of `temp` ($r \approx 0.99$). Categorical columns are cast to the pandas `category` dtype so XGBoost can split on them natively. The target is the raw hourly count `cnt`, and the Poisson loss with a log link (Section 4.2) absorbs its right skew (skewness 1.28). The split is day-level: whole days are assigned to training (70% of the 731 days) or test (30%) with `RANDOM_STATE = 42`, so neighboring hours of the same day never straddle the split. Randomizing over days suits a study whose unit of analysis is the explanation, not the forecast. A strictly temporal split would end the training window on 2012-05-27, so the `yr` effect would be learned from January to May 2012 only. That split is reported as a robustness check (Section 5.1). Nine features remain: `hr`, `temp`, `yr`, `weathersit`, `mnth`, `weekday`, `hum`, `windspeed` and `holiday`, with a test set of $n = 5\,227$ observations.

4.2 Models

XGBoost (XGB) (Chen & Guestrin 2016), a gradient-boosted tree ensemble, is used with `enable_categorical=True`, so categorical features need no one-hot encoding. The Explainable Boosting Machine (Lou et al. 2013, Nori et al. 2019) follows Equation (1) with 15 pairwise interaction terms f_{jk} (Section 4.4). Two losses were tried on each model, squared error and Poisson deviance. The loss was chosen by test accuracy and by the physical plausibility of the predictions.

Squared error is rejected because it produces negative predictions (133 on XGBoost, 411 on the EBM), which is impossible for count data. With Poisson deviance the XGBoost model achieves RMSE 45.44, MAE 27.00, $R^2 = 0.935$ and Poisson deviance 9.38. The EBM achieves RMSE 48.20, MAE 28.20, $R^2 = 0.927$ and Poisson deviance 10.81. The EBM is slightly weaker on all four metrics, the expected price of interpretability.

Contributions are in log space: a contribution of 0.2 multiplies the predicted rentals by about 1.22. Both generation prompts state this scale (Appendix A) but give no conversion formula, so the plain-language translation is left to the LLM. Neither instrument checks magnitude statements, since the rubric scores only direction, rank and structure, and the judge does not see `effect_range`.

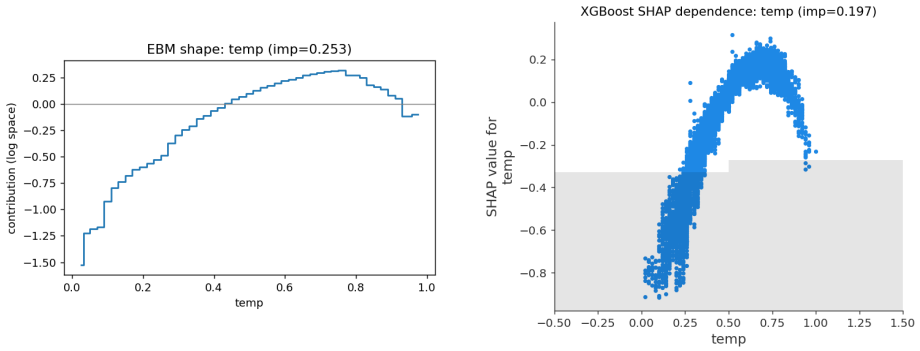


Figure 1. Per-feature global effect artifacts for `temp` on both XAI arms. These are the exact images the `vision` pipeline (and `get_feature_plot` in `tooluse`) receives. Left: the EBM shape function $f_{\texttt{temp}}$ read directly from `explain_global()`, a clean grid of bin midpoints against per-bin log-space contributions, with the feature importance printed in the title. Right: the XGBoost SHAP dependence scatter for the same feature, one point per training row (roughly 12 000 points), with SHAP's default gray feature-distribution histogram in the background and the importance printed in the title (`imp=0.197`). The plot is single-color, and the x-axis extends beyond the normalized range of `temp`. The two artifacts encode the same effect but with structurally different per-instance densities, and the aggregated form-type classifier of Section 3.4 operates on both.

4.3 Global Explanation Artifacts

For each of the 9 features and each of the two XAI models, two artifacts are produced. The first is a per-feature curve. For the EBM this is the learned shape function f_j read directly from `explain_global()`: a clean grid of bin midpoints on the horizontal axis and per-bin log-space contributions on the vertical axis. For XGBoost this is the SHAP dependence scatter for that feature: one point per training row, with the feature value on the horizontal axis and the TreeSHAP value on the vertical axis, roughly 12 000 points per feature. The scatter is kept as-is for the per-feature G2a JSON handover so the LLM reads the raw XGBoost output, and is averaged per distinct feature value only where a downstream consumer needs a curve: the ground-truth classifier, the deterministic template baseline, and the `json_all` and `tooluse_all` whole-model payloads.

The second artifact is a beeswarm. The XGBoost beeswarm is the standard SHAP beeswarm. The EBM beeswarm is constructed and is the subject of the next subsection. Figure 1 places one example of each per-feature curve side by side for the `temp` feature, so the difference in artifact structure that Section 4.4 exploits is visible at a glance.

4.4 Constructed EBM Beeswarm

An EBM is additive by Equation (1). Each per-instance main-effect contribution $f_j(x_{ij})$ is that shape function evaluated at the data point, the model's exact internal component rather than a post-hoc approximation. For an additive model without interactions, interventional SHAP with the training data as background reduces algebraically to these centered term contributions. With interactions, a feature's SHAP value would also re-

ceive a share of every interaction term it takes part in, so the constructed values match only the main-effect part of interventional SHAP. The XGBoost arm uses TreeSHAP's tree-path-dependent value function instead (Section 2). Both arms therefore share the scale (log space) and the importance measure (mean absolute contribution), but not an identical value function, and the constructed plot is a way to hand the EBM over with a familiar artifact, not an independent attribution method.

> The contributions are native and exact. Only the plot is a reconstruction. For XGBoost the situation is inverted: the beeswarm plot is standard, but the contributions are post-hoc via TreeSHAP.

Interaction terms are excluded so that the matrix has one column per feature and can be plotted like a SHAP matrix (implementation details in Appendix A). This is the main limitation of the constructed artifact. The 15 pairwise interactions carry 31% of the model's total term importance, and `hr&weekday` is its second most important term (importance 0.35, ahead of `temp` and `yr`), so the constructed beeswarm and the per-feature shape plots show only the main-effect part of what the EBM does.

The construction is validated twice. Its row ordering matches the EBM's own global importance ordering exactly (Spearman 1.000, a quasi-tautological check since both sides read the same term contributions), and it agrees with the XGBoost SHAP beeswarm at Spearman 0.883 with 100% direction agreement across the nine features. Hsiao et al. (2026), Figure 13, show a related visualization that applies SHAP on top of the EBM and so keeps the interaction terms. The version used here is simpler and cheaper but shows only the main effects.

The two beeswarms should not be compared as plots: colormap, point rendering and horizontal range differ for reasons unrelated to the models (Figure 2), so any cross-arm difference in the `vision_beeswarm` condition is partly rendering.

4.5 The Four Pipelines and the Tool Catalog

All three LLM handover formats share a system-prompt core and differ only in the handover block. The generator is Claude Sonnet 4.6 at $\mathrm{temperature} = 1.0$. Each answer starts with a short `<analysis>` scratchpad (Wei et al. 2022), which is stripped before scoring (Appendix A), followed by three bracketed sections: `[EFFECT]`, `[IMPORTANCE]` and `[RECOMMENDATION]`. The whole-model prompt asks for one `[FEATURE: name]` block with `[EFFECT]` and `[IMPORTANCE]` per feature and a single `[RECOMMENDATION]`, and each answer is split into nine records for scoring.

The `template` pipeline is a deterministic Python function that takes direction, form, peak, rank and importance from the same derivation as the ground truth (`describe_curve`, which calls `classify_shape`) and writes them out in fixed phrases. It is the RQ2 baseline. Because it reads the ground truth rather than the artifact, it cannot misread anything. It can only lose points through what its phrase map omits or words badly. It names no lowest categories, no sign change and no `note`, and it describes every categorical feature as "non-monotonic" with a contribution that "rises and falls across the range". The `json` pipeline hands the LLM the feature's curve as a JSON payload (`feature, kind, curve.x, curve.y, importance,`

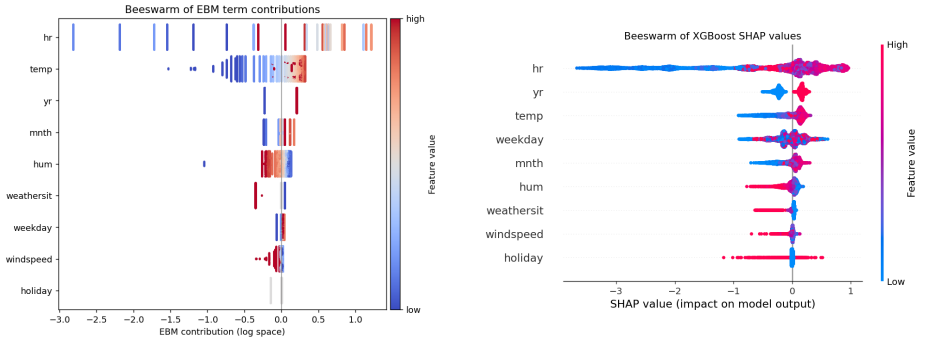


Figure 2. Beeswarms on both XAI arms. Left: the constructed EBM beeswarm from the model's exact additive term contributions (Section 4.4). Right: the standard XGBoost SHAP beeswarm from TreeSHAP. Both have row order = mean |contribution|, horizontal position = per-instance contribution in log space, color = feature value. Both carry a descriptive title, neither prints importance values. The three plot-level differences named in the text (colormap, point rendering, horizontal range) make cross-arm comparison of the images themselves misleading. The ordering and direction claims of Section 4.4 are what the two beeswarms have in common.

Table 3. Tool catalog for the pull condition. The last column is the number of G2a records ($n = 18$) in which the tool was called at least once. The near-perfect coverage of `get_feature_importances`, `get_feature_curve` and `get_feature_plot` carries the finding in Section 5.7: the pull condition stereotypes into a fixed three-tool sequence.

Tool	Purpose	used in G2a
`get_feature_importances`	full ranking, one row per feature	18 / 18
`get_feature_curve`	one feature's shape values	18 / 18
`get_feature_plot`	one feature's shape / dependence PNG	16 / 18
`get_target_overview`	summary of the target variable	3 / 18
`get_beeswarm_plot`	ranked view of all features	0 / 18

`rank, n_features`, for XGBoost the raw scatter) and asks it to read direction, shape, peak and importance strictly from those fields. The `vision` pipeline hands the LLM the feature's shape or dependence PNG and asks it to read direction, shape and peak visually. The user message also states the feature's rank in text ("Its global importance rank is r of 9", `prompts/global_feature.md`), so rank is available to the vision pipeline but not from the image. The `tooluse` pipeline is an agentic loop (Yao et al. 2023): nothing is pushed up front, and the model must call tools to retrieve what it needs before answering. The tool catalog is deliberately small (Table 3) so that any effect can be attributed to what the tools return, not to a large action space.

The `get_feature_plot` tool returns exactly the same PNG that the `vision` condition sees for the same feature. Pull and push therefore share the same artifact. Only the retrieval mechanism differs. Pull can therefore reach everything push delivers, and more. A pull deficit would be informative about the mechanism. A pull advantage could reflect information rather than mechanism.

Table 4. Prediction accuracy of both models on the day-level random split (used throughout) and on the strictly temporal split (reported here as a robustness check only).

Model	Split	Train N	Test N	RMSE	MAE	R^2
XGBoost	Random (day-level)	12 152	5 227	45.44	27.00	0.935
XGBoost	Strictly temporal	12 165	5 214	85.08	55.70	0.851
EBM	Random (day-level)	12 152	5 227	48.20	28.20	0.927
EBM	Strictly temporal	12 165	5 214	72.59	45.94	0.892

4.6 Reproducibility

The full run is reproducible from the public repository `https://github.com/antonpstahl/llm-assisted-xai-explanations`, which holds the code, the prompts and all raw results in the submitted state. Model identifiers and the retrieval date are pinned, `RANDOM_STATE = 42` threads through the split and the model fits, and a `RUN_API` flag replaces every LLM call with a deterministic stub, so the pipeline runs end-to-end offline. The test suite (350 tests on Python 3.13) passes on this state, and `analyses/numbers_ledger.py` recomputes the evaluation numbers of Sections 5.3–5.7 from `results/`. Appendix A describes the drift and truncation guards.

5 Results

5.1 Model Performance

Table 4 reports the prediction accuracy of both models. On the day-level random split both are strong ($R^2 \geq 0.927$). Refitted on the strictly temporal split, both degrade, XGBoost more than the EBM, because the second half of 2012 is unseen at fit time. The model gap is not the contribution of this paper. It is reported to underwrite the ground truth (Section 5.2).

5.2 Ground Truth Verification

All 18 references reproduce mechanically: the form, direction, monotonicity, importance rank and top categories re-derive exactly from the trained model with no manual corrections. Threshold sensitivity is a stricter test: the two derivation thresholds (`flat_importance`, `mono_tol`) are perturbed across the grid $\{0.015, 0.02, 0.025, 0.03, 0.035\} \times \{0.10, 0.15, 0.20, 0.25\}$, and the form labels are re-derived at every point. 15 of 18 labels are stable across the whole grid. The three boundary cases are `ebm/temp` (non-monotonic↔monotonic under a wider tolerance), `ebm/weekday` (a categorical feature whose importance of 0.024 sits just below the near-flat threshold of 0.025, so a lower threshold labels it categorical), and `xgb/windspeed` (near-flat↔non-monotonic at a lower flat threshold). Two of the three sit at the near-flat threshold, one at the monotonicity tolerance. The near-flat

Table 5. Per-feature descriptions (G2a). Rubric total is the mean of the three sub-scores in $[0, 1]$. Judge faithfulness is on a 1–5 scale (Section 3.5). Averaged over 9 features $\times$ 2 XAI models. Judge means carry 95% bootstrap confidence intervals (2 000 resamples over the 18 pairs, seed $= 42$). Both vendors are reported side by side: on the Anthropic judge no pairwise difference is significant. On the OpenAI judge the template vs. `json` contrast is Cliff's $\delta = 0.57$ (large) at raw $p = 0.005$ (Holm-adjusted $p_{\text{adj}} = 0.030$), sensitive to a single verdict.

Pipeline	Rubric total	Judge faith. (Anthropic)	Judge faith. (OpenAI)
`template`	0.944	4.50 [4.17, 4.83]	4.72 [4.50, 4.89]
`json`	0.889	4.28 [3.72, 4.72]	3.44 [2.78, 4.06]
`vision`	0.907	4.61 [4.22, 4.94]	3.83 [3.28, 4.33]
`tooluse`	0.880	4.44 [3.94, 4.89]	3.83 [3.17, 4.44]

Table 6. G2a stratified by ground-truth form type. All four generation pipelines pooled (n counted across formats). The near-flat stratum is where the failure mode lives. The monotonic stratum is at ceiling.

Form type	n	Judge faithfulness
Near-flat	20	3.85
Categorical	28	4.54
Non-monotonic	16	4.81
Monotonic	8	5.00

threshold is exactly where Section 5.3 finds the clearest pattern, which partly explains that pattern and partly limits it.

5.3 Per-Feature Descriptions (G2a)

Table 5 reports the per-feature comparison. On the rubric the template leads nominally (0.944). On the Anthropic judge it is second (4.50) behind `vision` (4.61), and no pairwise comparison is significant (Holm $p_{\text{adj}} \geq 0.38$, five of six Cliff's δ negligible). On the OpenAI judge the template leads by 0.89–1.28. Template vs. `json` reaches $\delta = 0.57$ (large) at raw $p = 0.005$ and Holm $p_{\text{adj}} = 0.030$, the only contrast on either judge that survives correction. It is fragile: with one verdict changed (`json_ebm_weathersit` scored 5, as the Anthropic judge scores it, instead of 3) the Holm value is 0.067, and corrected across both judges it is 0.061. Template vs. `vision` ($\delta = 0.47$, $p_{\text{adj}} = 0.11$) and vs. `tooluse` ($\delta = 0.33$, $p_{\text{adj}} = 0.14$) do not survive. With only 4–9 informative pairs on the Anthropic judge, its null is not equivalence, and Section 5.4 shows that a single draw cannot resolve small modality effects in any case.

The more informative pattern appears once the records are stratified by the ground-truth form of the feature (Table 6, all four handovers pooled, Anthropic judge). Judge faithfulness rises from 3.85 on near-flat features through 4.54 (categorical) and 4.81 (non-monotonic) to a saturated 5.00 on monotonic features, the direction expected from the hallucination literature (Ji et al. 2024, Maynez et al. 2020). On the near-flat stratum the template leads all three LLM formats on the rubric (0.933 against 0.667) and on

Table 7. Cross-vendor comparison on the near-flat stratum only, per generation pipeline. The template leads all three LLM formats on both judges. The gap is 0.8–1.4 points on the Anthropic judge and 1.6–2.4 on the OpenAI judge. The template shares the classifier with the ground truth and applies a fixed "negligible" framing when the feature is classified near-flat. The LLM formats have to recover that framing from a decisive-looking shape function (Figure 3).

Pipeline	near-flat mean (Anthropic)	near-flat mean (OpenAI)	n
`template`	4.60	4.80	5
`json`	3.20	2.40	5
`vision`	3.80	3.00	5
`tooluse`	3.80	3.20	5

Table 8. Judge faithfulness by ground-truth form: template vs. the mean of the three LLM formats, per judge. The near-flat template lead appears on both judges. The lower OpenAI scores for LLM answers appear in every stratum.

		Anthropic			OpenAI		
Form	n	template	LLM	gap	template	LLM	gap
Near-flat	5	4.60	3.60	+1.00	4.80	2.87	+1.93
Categorical	7	4.00	4.71	−0.71	4.71	3.95	+0.76
Non-monotonic	4	5.00	4.75	+0.25	5.00	3.83	+1.17
Monotonic	2	5.00	5.00	0.00	4.00	4.67	−0.67
All	18	4.50	4.44	+0.06	4.72	3.70	+1.02

both judges (Table 7), so this lead meets the convergence rule of Section 3.5. It remains descriptive, since with $n = 5$ pairs per stratum no test can reach significance, and it is largely an oracle advantage.

The template shares the classifier with the ground truth and therefore knows which features are near-flat. Its phrase map renders them as "negligible (roughly flat)", whereas the LLM per-feature prompt carries no such rule. The "must be called negligible" rule appears only in the whole-model prompt (rule 6), so the LLM formats have to recover the framing from a decisive-looking shape function (Figure 3). The template's near-flat advantage is therefore an oracle artifact, not evidence that it describes better. On the Anthropic judge this lead is canceled by a deficit on categorical features (4.00 against 4.71 for the LLM formats, Table 8). That deficit comes from the template's phrasing, not its content: it calls every categorical feature "non-monotonic", and the Anthropic judge deducts for exactly this (`template_ebm_weathersit` scores 3/5). The Anthropic-judge null of Table 5 therefore partly reflects a phrasing flaw in the baseline. What differs between the judges is a separate offset: the OpenAI judge scores the LLM answers 0.7–0.9 points lower in every stratum except the monotonic one, which has only two features (Section 5.6).

Figure 3 shows why near-flat features are hard. Both `windspeed` panels look decisive on their own auto-scaled y-axis, and only the importance printed in the title reveals that the effect is small. The XGB vision answer reads that cue. It states rank 8 of

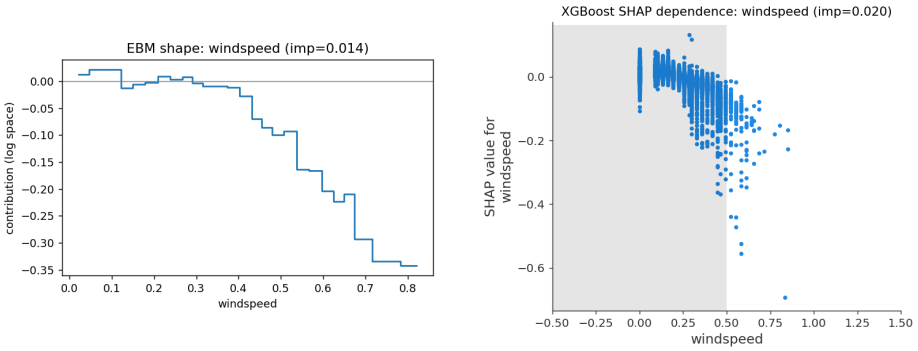


Figure 3. Per-feature vision inputs for `windspeed` on both XAI arms, exactly as received by the `vision` pipeline (and by the `get_feature_plot` tool). Each panel has its own auto-scaled y-axis. Left: EBM shape function, declining to about -0.35, the title prints `imp=0.014` (classified near-flat by the aggregated-importance rule, rank 8 of 9 on the EBM ordering, one place above `holiday`). Right: XGB dependence scatter: the title prints `imp=0.020`, a downward trend to about -0.2, with a single observation at -0.7. The contribution range here (≈ 0.36 on EBM) is about a fifth of what Figure 1 shows for `temp` (≈ 1.85), but the auto-scaled panels do not surface that comparison.

Table 9. Descriptive interaction between XAI method and handover format (G2a). Δ = EBM − XGB per handover, on the rubric total and on judge faithfulness for both judge vendors. Read as a pattern. No significance test is run because of the method/model confound.

	`template`	`json`	`vision`	`tooluse`	spread
Δ rubric total	0.000	−0.037	−0.074	+0.019	0.093
Δ judge faithfulness (Anthropic)	−0.111	+0.333	+0.111	+0.444	0.556
Δ judge faithfulness (OpenAI)	−0.111	+0.667	+0.778	+1.444	1.556

9 and an importance of 0.020, yet still describes a substantial drop in rentals at high wind: the importance reaches the `[IMPORTANCE]` section but not the `[EFFECT]` section.

Table 9 answers the interaction part of RQ5 descriptively. On the rubric total the EBM–XGB deltas spread by only 0.093 across the four formats. On judge faithfulness the three LLM formats do better on the EBM basis while the template does slightly better on XGB (spread 0.556 on the Anthropic and 1.556 on the OpenAI judge). Because XAI method is confounded with prediction model, this is a pattern, not an effect. Completeness is at ceiling (all cells 5.0). Across all 72 records, rubric and judge correlate at Spearman $\rho = 0.458$ (OpenAI 0.386), which is internal consistency between two ground-truth-bound scorers rather than independent validity.

5.4 Generation Variance

Table 5 rests on one draw per cell, so a variance study redrew 24 cells: three low-importance features (`windspeed`, `holiday` and `weekday`, near-flat in five of the six

Table 10. Generation variance vs. modality effect (G2a). Within-cell standard deviation is the mean of the standard deviations across the near-flat cells with rubric headroom (single-cell scale). Modality span is the max–min across the four format means in Table 5 (mean-of-18 scale). Both the rubric and the OpenAI rows quote the four-format span and, in the second value, the three-LLM-format span actually present in the variance study. The Anthropic span is 0.333 either way, because the template sits in the middle. The ratios are given for scale-visibility only, not for the interpretation, see the Monte Carlo below.

Instrument	Within-cell SD	Modality span	Ratio
Rubric total	0.089	0.064 / 0.027	1.4× / 3.3×
Judge faithfulness (Anthropic)	0.604	0.333	1.8×
Judge faithfulness (OpenAI)	0.553	1.28 / 0.39	0.4× / 1.4×

feature–model cells) plus `hr` as a ceiling control, on both XAI models and the three LLM formats. Two further draws per cell give 48 new generations, scored by the rubric and both judges. One draw reached the 2 048-token cap and is excluded, leaving 71. Table 10 compares the within-cell standard deviation, a single-cell scale, with the modality span, a mean-of-18 scale. The two are not directly comparable, and because the cells concentrate on the low-scoring stratum, the SDs are biased upward. The Monte Carlo below carries the interpretation.

Resampling one of the three draws per cell (5 000 iterations, seed $= 42$) and averaging the eight cells per format produces all six possible orderings of the three LLM formats, each in 1.6–26.6% of iterations. A second Monte Carlo holds the ten non-redrawn cells fixed and resamples the eight redrawn ones, which puts the result on the full mean-of-18 scale. The resampled format means (`vision` 4.48, `tooluse` 4.41, `json` 4.20) have standard deviations of 0.08–0.10, about a quarter of the observed span. That is enough to swap the two top formats but not to lift `json` off the bottom. Appendix F gives the tie rules and shares of both runs. Because both runs resample from only three draws per cell, they understate rather than overstate the variability:

> The modality ordering across the three LLM handover formats read off single draws is not interpretable at the top. Resampling the same instrument at the same $\mathrm{temperature} = 1.0$, `vision` finishes strictly above `tooluse` in 66% of full-18 iterations, strictly below it in 19% and level with it in 15%. `json` is strictly below both other formats in 89%.

The ceiling control confirms this reading. `hr` is at ceiling on every LLM format, and its within-cell SD is 0.000 on the rubric and on the Anthropic judge, so the variance sits in the low-importance cells and the SDs do not generalize beyond them. The OpenAI judge shows an SD of 0.385 even on these control cells, so part of its spread is judge noise. Only the rubric SD (0.089) is pure generation variance, and it is 3.3 times the span between the three LLM formats (Table 10).

Table 11. Whole-model descriptions (G2b), Axis 1: per ground-truth field. The rank stated column counts how many of the 18 descriptions per condition committed to a rank at all. Where a rank was stated, it was exactly correct in every condition (90/90 across all five: $18 + 18 + 18 + 18 + 18$). Structure is shown for the beeswarm conditions but is not part of their `fair_total`.

Condition	direction	rank	rank stated	structure	fair_total
`json_all`	0.986	1.000	18 / 18	0.917	0.968
`vision_all`	0.972	1.000	18 / 18	0.750	0.907
`tooluse_all`	0.986	1.000	18 / 18	0.861	0.949
`json_beeswarm`	0.972	1.000	18 / 18	0.556	0.986
`vision_beeswarm`	0.889	1.000	18 / 18	0.611	0.944

5.5 Whole-Model Descriptions (G2b)

Coverage is 9/9 in all ten conditions, so it is a method check, not a result. Every G2b cell is a single generation. The ten whole-model answers (5 conditions $\times$ 2 XAI arms) are split into 90 records, so each 18-record row in Tables 11 and 12 is a slice of two draws, not 18 independent observations, and the single-draw fragility of Section 5.4 attaches to every G2b number.

Table 11 reports Axis 1 per ground-truth field on the rubric. Every condition commits to a rank for every feature, and every stated rank is exactly correct (90/90). The rubric's rank parser accepts the many phrasings the generator uses and is locked in by a dedicated test suite (Appendix C). Rank therefore does not discriminate between conditions, and the differences sit in the direction and structure fields.

`vision_all` is the only whole-model condition without an explicit numeric rank channel (Table 2). It still states all nine ranks correctly on both arms (`fair_total` 0.926 on EBM, 0.889 on XGB), and the answers quote the importance printed in each plot title (XGB `holiday`: "least important feature in the model (importance 0.012)"). Image order carries the same ordering, so the design cannot fully separate the two channels, but the quotes point to the titles. The prompt also suggests judging importance from the vertical range of the plots. That proxy works on the EBM arm (Spearman $\rho = 0.90$ with importance) but not on the XGB arm ($\rho = 0.23$), where rare extreme SHAP values make the `holiday` scatter the second widest although `holiday` ranks last. The XGB answers follow the printed values, as prompt rule 5 asks, which motivates Implication 1 of Section 6.2. The remaining gap to `json_all` on XGB (0.889 against 0.963) sits in the structure field (0.722 against 0.889), not in rank: the nine-plot handover loses shape detail, not the ranking.

Axis 2 contrasts push (`json_all`, `vision_all`) with pull (`tooluse_all`). Pull and `json_all` tie on EBM, `json_all` leads by 0.037 on XGB, and pull leads `vision_all` by 0.046 (EBM) and 0.037 (XGB, Table 12). No push–pull difference exceeds 0.05, so "pull beats numeric push" is not supported. Each gap rests on one or two sub-scores of single features in a single draw, and in the variance study the rubric total changed between draws in 11 of the 18 redrawn non-control cells, so gaps of this size are within what a redraw produces.

Table 12. Axis 2 (mechanism) disaggregated. Each row contrasts pull (`tooluse_all`) with one push condition on one XAI arm. A negative delta means the push condition beat pull. No push–pull difference exceeds 0.05, `json_all` ties pull on EBM and leads it by 0.037 on XGB.

Push condition	XAI model	`fair_total` pull	`fair_total` push	Δ (pull − push)
`json_all`	EBM	0.972	0.972	0.000
`json_all`	XGB	0.926	0.963	−0.037
`vision_all`	EBM	0.972	0.926	+0.046
`vision_all`	XGB	0.926	0.889	+0.037

Table 13. Cross-vendor Krippendorff's α per criterion (Anthropic vs. OpenAI). Interval metric. All G2a values are below the 0.667 tentative threshold and completeness on G2a is undefined: both judges give 5.0 on all 72 records.

Criterion	G2a α	G2b α
Faithfulness	0.481	0.295
Clarity	0.435	0.396
Completeness	n/a	0.638

The beeswarm pair is the only Axis 1 contrast with matched information. On the rubric `json_beeswarm` scores 1.000 / 0.972 and `vision_beeswarm` 0.917 / 0.972 (EBM / XGB), a small gap on the EBM direction sub-score and a tie on XGB. The model reads the swarm image essentially as well as the equivalent numbers, against the expectation from chart-understanding benchmarks (Wang et al. 2024) that the image would be read worse.

The judges score every split record against the full reference, including form, peak and sign change, so the beeswarm conditions (2.6–4.0) are lowered by construction and comparable only with each other. Both judges rate `vision_all_xgb` below `vision_all_ebm` (Anthropic 4.22 vs. 5.00, OpenAI 2.78 vs. 4.11), consistent with the structure gap on the rubric. They split on the beeswarm pair: the OpenAI judge favors the numbers (3.44 vs. 2.83), the Anthropic judge the image (3.94 vs. 3.61). Both place pull slightly ahead of `json_all` (+0.11 on each). With one draw per cell and a G2b faithfulness α of 0.295 (Table 13), these residuals support no judge-side conclusion, so the beeswarm and pull statements above remain rubric-instrument statements.

5.6 Judge Robustness

Table 13 reports the cross-vendor Krippendorff's α. All values sit below the 0.667 threshold usually cited for tentative reliability. Completeness on G2a is undefined because both judges give 5.0 on all 72 records, and on the more demanding G2b track it rises to 0.638.

The vendor gap is not a uniform strictness shift. The OpenAI judge scores the three LLM formats 0.61–0.83 lower than the Anthropic judge, but the template 0.22 higher (4.72 vs. 4.50). The gap opens where a Claude model wrote the answer and reverses for

Table 14. Per-record process and cost metrics on G2a (mean across 18 records per format).

Pipeline	Tokens in	Tokens out	Latency (s)	Tool calls
`template`	0	0	0.0	n/a
`json`	82 421	720	18.2	n/a
`vision`	1 554	704	17.8	n/a
`tooluse`	69 228	901	25.0	3.1

Table 15. Input tokens per record, split by XAI model. The `json` and `tooluse` handovers ship almost two orders of magnitude more on XGB than on EBM.

Pipeline	Tokens (EBM)	Tokens (XGB)	Ratio
`json`	483	164 360	340×
`tooluse`	1 887	136 569	72×
`vision`	1 469	1 640	1.1×

the deterministic template, partly because the Anthropic judge deducts for the template's categorical phrasing (Section 5.3). This is consistent with the self-preference bias of Panickssery et al. (2024), since generator and Anthropic judge share a model family (not a model). Two other explanations fit as well. The deduction rule punishes extra or over-committed claims, which the LLM answers make and the template does not, and the OpenAI judge may apply deductions more readily. The judges also differ in model generation (`gpt-4o` is older than `claude-opus-4-8`) and in decoding (temperature 0 vs. the API default). The three explanations are not identifiable on this data, which is why Table 5 reports both vendors side by side.

5.7 Process and Cost Metrics

Table 14 reports mean tokens, latency and tool calls per G2a record. The template costs nothing to run.

The pooled token figures hide an asymmetry (Table 15). The XGB dependence scatter ships 12 152 raw points per feature, so the `json` payload is 340× larger on XGB than on EBM, where it is smaller than the image. This confounds the RQ5 columns of Table 9. The asymmetry is defensible at the level of content, since the plot shows all 12 152 points too, but not at the level of tokens.

The per-feature pull condition does not behave as a flexible retrieval loop. In 16 of 18 records the agent calls `get_feature_importances` → `get_feature_curve` → `get_feature_plot`, and it retrieves the importance in every record although the user message already states the rank. This sequence is the union of what the `json` and `vision` handovers push, a plausible mechanism for the tie between the three LLM formats in Table 5. Tool calls grow with the size of the request (3.1 per G2a record, 16–17 per whole-model answer, Appendix A) rather than with new retrieval strategies.

5.8 Qualitative Error Analysis

Manual inspection of the G2a records scoring ≤ 3 on judge faithfulness gives the third, non-numeric view. On the Anthropic judge 12 records fall in that pool, 8 of them near-flat, 3 categorical and 1 non-monotonic. Three are template records that lose points for calling a categorical feature "non-monotonic". On the OpenAI judge the 20 low records spread more evenly (8 near-flat, 7 categorical, 4 non-monotonic, 1 monotonic), so the near-flat concentration holds on the Anthropic judge only. Excerpts are in `analyses/error_examples_by_formtype.md`.

The picture is more nuanced than over-attribution. On `xgb/windspeed`, both low-scoring records describe a consistent decline and miss the small positive peak near 11 km/h and the sign change near 16 km/h that both judges cite.[1] The vision record quotes the importance from the plot title and still describes a substantial drop, so the missing framing rule is not the only cause. That XGB scatters absorb tree interactions into per-instance heterogeneity could explain part of this, but the reading is speculative and the effect is small.

On `holiday`, `json_xgb_holiday` states that "holidays mostly reduce demand (likely by eliminating commuter trips)". The per-instance SHAP values are indeed strongly negative in the commute hours and near zero at midday, but the JSON handover contained no hour information, so the claim is a world-knowledge guess that happens to match the model: faithful to the model, but not grounded in the handover. The Anthropic judge scored it 5/5 and the OpenAI judge 1/5. The aggregated ground truth conflates "low importance" with "flat curve" and can penalize descriptions that surface such genuine heterogeneity (Section 6.4).

6 Discussion

6.1 Answering the Research Questions

RQ1: Representation. A single beeswarm carries rank and direction but not shape (structure 0.56–0.61 against 0.75–0.92 for the nine-plot conditions). At matched information the LLM reads the beeswarm image about as well as the equivalent numbers, a rubric-instrument statement since the two judges split on this pair (Section 5.5). A collection of plots recovers the ranking when each plot carries its importance in the title. Compared with the large gaps on realistic charts that Wang et al. (2024) report, the dense beeswarm is read imperfectly, but it is read.

RQ2: Baseline. The answer depends on the judge. On the Anthropic judge the deterministic template is indistinguishable from the three LLM handovers, partly because that judge penalizes the template's phrasing of categorical features. On the OpenAI judge the template leads, but only the template–`json` contrast survives Holm correction, and it rests on a single verdict (Section 5.3). Most of the template's lead sits on near-flat

[1] Wind speed is de-normalized as value $\times$ 67, as in the generation prompts. The dataset documentation gives no unit, and km/h is assumed by analogy with the Celsius temperature scale of the same file.

features, where it profits from sharing the classifier with the ground truth. At small task size a template is therefore a strong baseline. Whether the LLM adds value on whole-model descriptions was not measured, since no whole-model template was run.

RQ3: Handover format. On the per-feature task the three LLM handovers are not separable: no pairwise difference survives correction on either judge, and the ordering read off a single draw is not interpretable (Section 5.4). Only `json` in last place is stable. The null is therefore a limit of single-draw designs, not only a failure to find an effect.

RQ4: Mechanism. Pull shows no measurable advantage over numeric push at whole-model complexity. On the rubric the two lie within 0.05 on both XAI models, while both judges place pull slightly ahead ($+0.11$), all from one draw per arm (Section 5.5). The pull loop holds at least the numeric-push information and follows the tools its prompt names (Section 5.7), so the near-tie is what one would expect if the retrieval mechanism itself adds nothing.

RQ5: XAI method. Because XAI method is confounded with the prediction model, the answer is descriptive. The rubric shows no interaction with the handover format. On the judges the LLM handovers do slightly better on the EBM basis and the template on XGB (Table 9). The token asymmetry does not explain this alone, since `vision` shows the same direction at a token ratio of only $1.1\times$. A related caveat is that the EBM's interaction terms carry 31% of its importance but are missing from the plots handed over (Section 4.4), which is consistent with EBM `weekday` being labeled near-flat while its XGB counterpart ranks fourth.

6.2 Implications

For research, the handover channel is a design parameter of LLM translation layers in its own right. Its effect on faithfulness is smaller than the variance of a single draw, so single-draw and single-judge evaluations cannot rank such designs.

For practice, the implications are ordered by how well the results support them:

1. When a ranking has to be recovered from a collection of plots, print each plot's importance in its title or supply it separately. With titled plots every rank was recovered, whereas the plotted vertical range would have misled on the XGB arm ($\rho = 0.23$). An untitled condition was not run, so the size of this benefit is not measured.
2. For dense post-hoc artifacts, prefer the image handover on cost grounds. On the XGB arm it needs 1 640 instead of 164 360 input tokens per feature, and neither judge detects a faithfulness difference between the two.
3. Compare prompt or handover variants on several draws per cell, since redraws at the default temperature reorder the three LLM handovers (Section 5.4).
4. Do not rely on a single LLM judge. Two strong judges from different vendors agree only at $\alpha = 0.48$ (G2a) and 0.30 (G2b) on faithfulness and disagree on whether the template or `vision` leads.

5. Treat a deterministic template as the cost floor for narrow per-feature descriptions, and give the LLM the same framing rules. The "negligible" rule for near-flat features belongs in the per-feature prompt, not only in the whole-model prompt.
6. Separate "low importance" from "flat curve" in an aggregated ground truth, so that descriptions surfacing genuine per-instance heterogeneity are not penalized.

6.3 Contrast with Related Work

The most closely related paper is Lukassen, Herrmann, Weisser, Saefken & Kneib (2026), which varies XAI method, LLM and prompting and finds that the LLM choice dominates. Here the LLM is held constant and the handover channel varied, so this paper can neither confirm nor contest that finding, nor their report that chain-of-thought prompting lowered quality against the general expectation of Wei et al. (2022). The same group's follow-up (Lukassen, Herrmann, Weisser, Silbersdorff, Saefken & Kneib 2026) sharpens the convincing-versus-faithful gap of Section 1.1. With LLM judges standing in for non-expert readers, high-quality narratives inflate confidence without improving decision accuracy, echoing the false-trust warning of Slack et al. (2023). This is a direct argument for measuring faithfulness against a reference rather than reader perception.

The null of Kruschel et al. (2026) (Section 1.2) is between two assistance forms, not between assistance and no assistance. The present paper is consistent with that null from the explanation-quality side. Pull does not beat numeric push on the whole-model rubric, and on the per-feature task neither judge separates `tooluse` from `json` after correction. Whether the shared explanation limits user performance cannot be tested here.

Bordt et al. (2024) and Lengerich et al. (2023) describe glass-box models with an LLM, the latter already by decomposing them into per-feature steps, but neither measures the description against a reference. That evaluation is what this paper adds, together with a beeswarm built from the EBM's exact contributions (Section 4.4).

In line with Rudin (2019), the EBM side loses little accuracy on this dataset ($R^2 = 0.927$ vs. 0.935), and its per-instance contributions are exact rather than post-hoc. This is an argument for glass-box as a default when the accuracy cost is small, but it is descriptive here because the RQ5 confound applies.

6.4 Limitations

Several results are bounded by the design, and some plausible headlines are not supported. First, every G2b cell is a single whole-model generation, so the 9/9 rank commitment on `vision_all`, the pull $\approx$ push result and the beeswarm-pair residual are each one draw per XAI arm, and the Monte Carlo of Section 5.4 applies to them. Second, the judge rubric (`prompts/judge_system_global.md`) deducts for missing negligibility framing, which the per-feature generation prompt never asks for while the template applies it by construction. The near-flat gap between template and LLMs is therefore partly a fingerprint of this asymmetry (implication 5). Third, on G2a the judge sees the tool-use retrieval transcript but not the JSON or vision payloads, so the four per-feature handovers are not judged on symmetric evidence. The G2b split records carry no trace, so RQ4 is unaffected. Fourth, the aggregated ground truth conflates "low importance" with "flat

curve". On `holiday` the LLM's commuter claim matches the model's per-instance pattern, and on `xgb/windspeed` the deducted "consistently decreasing" descriptions miss only a $+0.018$ bump near 11 km/h, about 2% of demand (Section 5.8). Fifth, no whole-model template was run, so the LLM's added value on the whole-model task is a design expectation, not a measurement. Finally, the scope bounds generalization: one dataset, one generator model and 18 feature–model pairs, with faithfulness measured against a reference rather than usefulness for readers. Appendix G collects the full catalog.

6.5 Future Work

Several extensions follow. Repeated draws per cell, including the whole-model cells, would turn single-draw statements into measured effects. A cross-vendor comparison on the generator side would separate the LLM effect from the modality effect, and a user study would add usefulness to faithfulness. A second dataset would let RQ5 be run as a within-model contrast, and a mixed-effects model with feature as a random effect would sharpen the analysis. On the design side, varying image order and plot titles independently would separate the two rank channels of `vision_all`, a shared y-axis would test the role of per-plot autoscaling in near-flat over-attribution, and a `template_all` condition would give RQ2 a whole-model measurement.

The results also suggest a design direction beyond measurement. An LLM could state its claims in a fixed vocabulary whose truth conditions are checked against the glass-box model before release, including interaction terms and negligibility judged by the model's own confidence intervals.

7 Conclusion

This paper asked how the channel through which an XAI artifact reaches an LLM affects explanation faithfulness. It compared a deterministic template with JSON, vision and tool-use handovers on an XGBoost arm with TreeSHAP and an EBM arm, scoring 72 per-feature and ten whole-model descriptions against a mechanically derived ground truth with a deterministic rubric and judges from two vendors.

Four results stand, each tied to its track and instrument. First, on the per-feature task, redraws reorder the three LLM handovers, so a single draw cannot show which format leads. Second, on the whole-model task and the rubric, the LLM reads a beeswarm image about as well as the same numbers, but a beeswarm alone loses the shape detail of nine per-feature plots. Third, on the same task and instrument, pulling artifacts through tools does not beat pushing them as numbers. Fourth, whether a deterministic template matches the LLMs depends on the judge, and its one surviving advantage is fragile. In practice, an image handover of dense XGBoost plots needs a hundredth of the input tokens of the raw numbers without a detectable loss in faithfulness, and with cross-vendor agreement of only α 0.30–0.48, a single LLM judge is not a sufficient faithfulness check.

Alongside these results, the paper builds a beeswarm from an EBM's exact main-effect contributions, which lets a glass-box and a post-hoc arm be handed over with the same artifact. Open questions are user usefulness, effects across generators, repeated whole-model draws and a ground truth that separates low importance from flat shape.

References

Apley, D. W. & Zhu, J. (2019), 'Visualizing the effects of predictor variables in black box supervised learning models', arXiv preprint arXiv:1612.08468. Published in Journal of the Royal Statistical Society: Series B 82(4), pp. 1059–1086.

Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., Garcia, S., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R. & Herrera, F. (2020), 'Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI', *Information Fusion* **58**, 82–115.

Bauer, K., von Zahn, M. & Hinz, O. (2023), 'Expl(AI)ned: The impact of explainable artificial intelligence on users' information processing', *Information Systems Research* **34**(4), 1582–1602.

Bokstaller, J., Altheimer, J., Dormehl, J., Buss, A., Wiltfang, J., Schneider, J. & Röglinger, M. (2025), Enhancing ML model interpretability: Leveraging fine-tuned large language models for better understanding of AI, *in* 'Proceedings of the 33rd European Conference on Information Systems (ECIS 2025)'.

Bordt, S., Lengerich, B., Nori, H. & Caruana, R. (2024), 'Data science with LLMs and interpretable models', arXiv preprint arXiv:2402.14474. XAI4Sci: Explainable Machine Learning for Sciences, AAAI 2024 Workshop.

Brasse, J., Broder, H. R., Förster, M., Klier, M. & Sigler, I. (2023), 'Explainable artificial intelligence in information systems: A review of the status quo and future research directions', *Electronic Markets* **33**, 26.

Chen, T. & Guestrin, C. (2016), XGBoost: A scalable tree boosting system, *in* 'Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)', ACM, pp. 785–794.

Cliff, N. (1993), 'Dominance statistics: Ordinal analyses to answer ordinal questions', *Psychological Bulletin* **114**(3), 494–509.

European Union (2024), 'Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)', Official Journal of the European Union, OJ L, 2024/1689, 12.07.2024. `https://eur-lex.europa.eu/eli/reg/2024/1689/oj`. CELEX: 32024R1689 [Accessed: 9 September 2026].

Fanaee-T, H. & Gama, J. (2014), 'Event labeling combining ensemble detectors and background knowledge', *Progress in Artificial Intelligence* **2**(2–3), 113–127.

Friedman, J. H. (2001), 'Greedy function approximation: A gradient boosting machine', *The Annals of Statistics* **29**(5), 1189–1232.

Gregor, S. & Hevner, A. R. (2013), 'Positioning and presenting design science research for maximum impact', *MIS Quarterly* **37**(2), 337–355.

Hayes, A. F. & Krippendorff, K. (2007), 'Answering the call for a standard reliability measure for coding data', *Communication Methods and Measures* **1**(1), 77–89.

He, Y. & Martens, D. (2026), 'An agentic approach to generating XAI-narratives', arXiv preprint arXiv:2603.20003.

Hevner, A. R., March, S. T., Park, J. & Ram, S. (2004), 'Design science in information systems research', *MIS Quarterly* **28**(1), 75–105.

Holm, S. (1979), 'A simple sequentially rejective multiple test procedure', *Scandinavian Journal of Statistics* **6**(2), 65–70.

Hsiao, C.-H., Kumar, K. & Rathje, E. M. (2026), 'Domain-informed explainable boosting machines for trustworthy lateral spread predictions', arXiv preprint arXiv:2603.17175.

Ichmoukhamedov, T., Hinns, J. & Martens, D. (2024), 'How good is my story? Towards quantitative metrics for evaluating LLM-generated XAI narratives', arXiv preprint arXiv:2412.10220.

Jeck, J., Leiser, F., Hüsges, A. & Sunyaev, A. (2025), TELL-ME: Toward personalized explanations of large language models, *in* 'Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems (CHI EA '25)', ACM, Yokohama, Japan.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y., Chen, D., Dai, W., Chan, H. S., Madotto, A. & Fung, P. (2024), 'Survey of hallucination in natural language generation', arXiv preprint arXiv:2202.03629. Extended version (v7) of the article published in ACM Computing Surveys 55(12), pp. 1–38.

Kantharaj, S., Leong, R. T. K., Lin, X., Masry, A., Thakkar, M., Hoque, E. & Joty, S. (2022), 'Chart-to-Text: A large-scale benchmark for chart summarization', arXiv preprint arXiv:2203.06486. Published in Proceedings of ACL 2022 (Long Papers).

Kaur, H., Nori, H., Jenkins, S., Caruana, R., Wallach, H. & Wortman Vaughan, J. (2020), Interpreting interpretability: Understanding data scientists' use of interpretability tools for machine learning, *in* 'Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)', ACM, Honolulu, HI, USA, pp. 1–14.

Kruschel, S., Rosenberger, J., Bohlen, L., Kraus, M. & Zschech, P. (2026), Can conversational XAI improve user performance? An experimental study, *in* 'Proceedings of the 34th European Conference on Information Systems (ECIS 2026)'. Short Paper.

Lengerich, B. J., Bordt, S., Nori, H., Nunnally, M. E., Aphinyanaphongs, Y., Kellis, M. & Caruana, R. (2023), 'LLMs understand glass-box models, discover surprises, and suggest repairs', arXiv preprint arXiv:2308.01157.

Liu, Y., Iter, D., Xu, Y., Wang, S., Xu, R. & Zhu, C. (2023), 'G-Eval: NLG evaluation using GPT-4 with better human alignment', arXiv preprint arXiv:2303.16634. Published in Proceedings of EMNLP 2023.

Lou, Y., Caruana, R., Gehrke, J. & Hooker, G. (2013), Accurate intelligible models with pairwise interactions, *in* 'Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '13)', ACM, pp. 623–631.

Lukassen, F., Herrmann, J., Weisser, C., Saefken, B. & Kneib, T. (2026), 'From XAI to stories: A factorial study of LLM-generated explanation quality', arXiv preprint arXiv:2601.02224.

Lukassen, F., Herrmann, J., Weisser, C., Silbersdorff, A., Saefken, B. & Kneib, T. (2026), 'Quality without usefulness: LLM-generated XAI narratives as trust heuristics rather than decision aids', arXiv preprint arXiv:2605.26770.

Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N. & Lee, S.-I. (2019), 'Explainable AI for trees: From local explanations to global understanding', arXiv preprint arXiv:1905.04610.

Lundberg, S. M. & Lee, S.-I. (2017), A unified approach to interpreting model predictions, *in* 'Advances in Neural Information Processing Systems 30 (NIPS 2017)'. Cited from arXiv:1705.07874v2.

Martens, D., Hinns, J., Dams, C., Vergouwen, M. & Evgeniou, T. (2025), 'Tell me a story! Narrative-driven XAI with large language models', *Decision Support Systems* **191**, 114402.

Masry, A., Long, D. X., Tan, J. Q., Joty, S. & Hoque, E. (2022), 'ChartQA: A benchmark for question answering about charts with visual and logical reasoning', arXiv preprint arXiv:2203.10244. Published in Findings of the Association for Computational Linguistics: ACL 2022.

Maynez, J., Narayan, S., Bohnet, B. & McDonald, R. (2020), On faithfulness and factuality in abstractive summarization, *in* 'Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL 2020)', pp. 1906–1919.

Molnar, C. (2025), *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*, 3rd edn, Independently published. Available at `https://christophm.github.io/interpretable-ml-book/` [Accessed: 13 October 2026].

Nori, H., Jenkins, S., Koch, P. & Caruana, R. (2019), 'InterpretML: A unified framework for machine learning interpretability', arXiv preprint arXiv:1909.09223.

Panickssery, A., Bowman, S. R. & Feng, S. (2024), 'LLM evaluators recognize and favor their own generations', arXiv preprint arXiv:2404.13076.

Peffers, K., Tuunanen, T., Rothenberger, M. A. & Chatterjee, S. (2007), 'A design science research methodology for information systems research', *Journal of Management Information Systems* **24**(3), 45–77.

Romano, J., Kromrey, J. D., Coraggio, J. & Skowronek, J. (2006), Appropriate statistics for ordinal level data: Should we really be using t-test and Cohen's d for evaluating group differences on the NSSE and other surveys?, *in* 'Annual Meeting of the Florida Association of Institutional Research (FAIR)', Cocoa Beach, FL, USA. Conference presentation.

Rudin, C. (2019), 'Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead', arXiv preprint arXiv:1811.10154. Published in Nature Machine Intelligence 1, pp. 206–215.

Schulhoff, S., Ilie, M., Balepur, N., Kahadze, K., Liu, A., Si, C., Li, Y., Gupta, A., Han, H., Schulhoff, S., Dulepet, P. S., Vidyadhara, S., Ki, D., Agrawal, S., Pham, C., Kroiz, G., Li, F., Tao, H., Srivastava, A., Da Costa, H., Gupta, S., Rogers, M. L., Goncearenco, I., Sarli, G., Galynker, I., Peskoff, D., Carpuat, M., White, J., Anadkat, S., Hoyle, A. & Resnik, P. (2025), 'The Prompt Report: A systematic survey of prompt engineering techniques', arXiv preprint arXiv:2406.06608.

Singh, C., Askari, A., Caruana, R. & Gao, J. (2023), 'Augmenting interpretable models with large language models during training', *Nature Communications* **14**, 7913.

Slack, D., Krishna, S., Lakkaraju, H. & Singh, S. (2023), 'Explaining machine learning models with interactive natural language conversations using TalkToModel', *Nature Machine Intelligence* **5**(8), 873–883.

Spitzer, P., Celis, S., Martin, D., Kühl, N. & Satzger, G. (2024), Looking through the deep glasses: How large language models enhance explainability of deep learning models, *in* 'Proceedings of Mensch und Computer 2024 (MuC '24)', ACM, Karlsruhe, Germany, pp. 566–570.

Wang, Z., Xia, M., He, L., Chen, H., Liu, Y., Zhu, R., Liang, K., Wu, X., Liu, H., Malladi, S., Chevalier, A., Arora, S. & Chen, D. (2024), 'CharXiv: Charting gaps in realistic chart understanding in multimodal LLMs', arXiv preprint arXiv:2406.18521.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q. V. & Zhou, D. (2022), Chain-of-thought prompting elicits reasoning in large language models, *in* 'Advances in Neural Information Processing Systems 35 (NeurIPS 2022)'.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K. & Cao, Y. (2023), ReAct: Synergizing reasoning and acting in language models, *in* 'Proceedings of the 11th International Conference on Learning Representations (ICLR 2023)'.

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E. & Stoica, I. (2023), 'Judging LLM-as-a-judge with MT-Bench and Chatbot Arena', arXiv preprint arXiv:2306.05685. Published in Advances in Neural Information Processing Systems 36.

Zytek, A., Pidò, S. & Veeramachaneni, K. (2024), 'LLMs for XAI: Future directions for explaining explanations', arXiv preprint arXiv:2405.06064. CHI 2024 Workshop on Human-Centered Explainable AI.

A Prompts and Reproducibility

The pipeline prompts, verbatim, are collected in the repository under `prompts/`: the per-feature prompt in `prompts/global_feature.md`, the whole-model prompt in `prompts/global_whole.md`, the judge prompts in `prompts/judge_system.md` and `prompts/judge_system_global.md`. The SHA-256 hash gate in `tests/test_prompt_golden.py` covers the four local-track prompt files (`pipeline_04_json.md`, `pipeline_05_vision.md`, `pipeline_06_tooluse.md`, `judge_system.md`) only. The global-track prompts (`global_feature.md`, `global_whole.md`, `judge_system_global.md`) are not hashed. Their integrity is enforced instead by byte-identity checks on the shared prompt core at assembly time (`tests/test_global_feature.py`, `tests/test_global_whole.py`).

The template's shape-to-phrase map is in `notebooks/04Ga_Global_Template_Baseline.ipynb`. It renders a `near_flat` shape as "negligible (roughly flat)".

Scratchpad and splitting. A stripping helper (`utils/llm.py strip_scratchpad`) removes the scratchpad block (`<analysis>`, or `<thinking>` where the model uses that tag) before scoring, so the scorers see only the visible answer. A data-level test (`tests/test_no_scratchpad_leak.py`) checks that no persisted record carries a scratchpad block. `split_whole_model_record` cuts each whole-model answer into nine split records.

Constructed EBM beeswarm. The construction uses `model.eval_terms(X)`, the vectorized equivalent of the per-instance `explain_local(X)` term scores, and a byte-identical check was run on the main-effect terms. In Figure 2 the EBM points form piecewise-constant vertical bars because its contributions are per-bin (one value per shape-function bin), whereas SHAP's per-instance contributions form a kernel-density swarm. The colormaps also differ (coolwarm against SHAP's blue-to-magenta default).

Tool calls. In the per-feature records `get_feature_importances` is called first in 15 of 18 (the other three open with `get_target_overview`), and exactly the three-tool triple and nothing else occurs in 13 of 18. The local pre-study averages 6.2 tool calls per record. Both whole-model pull runs open with one call each to `get_target_overview`, `get_feature_importances` and `get_beeswarm_plot`, then call `get_feature_curve` for each of the nine features and finish with four (EBM) or five (XGB) `get_feature_plot` calls.

Drift guards. On the local-track prompts, a golden SHA-256 hash test refuses to run if any of the four pinned prompt files has drifted. On the global-track prompts, byte-identity of the shared core is asserted at prompt-assembly time. A README consistency test wraps every numeric table in the repository README in `<!-- AUTO-TABLE -->` sentinels and regenerates them from `results/`, and the corresponding pytest fails if the README contains a stale table. This is a local pre-commit check rather than a CI gate (the repository ships no CI pipeline). `tests/test_analyses_consistency.py` guards against drift between the analyses reports and `results/`, and `tests/test_rubric_rank.py` locks in the verbal-ordinal phrasings the rank parser must accept.

Truncation guards. `stop_reason` and `max_tokens` are persisted on every generation path, and an assertion refuses to write a truncated record, one whose `stop_reason` is `max_tokens` or whose output-token count reaches the cap. Records generated before these fields were added (the per-feature `json` records, the EBM `vision` records, six of the ten whole-model records and half of the variance draws) carry only the output-token count, which the evaluation code uses as a fallback to detect and exclude one truncated draw (`json_xgb_weekday_gen1`, Section 5.4).

B Ground-Truth Schema

Each ground-truth record has the fields `form` (one of `near-flat`, `monotonic`, `non-monotonic`, `categorical`), `direction` (one of `increasing`, `decreasing`, `mixed`, `categorical`), `monotonicity` (a string: `monotonic_increasing`, `monotonic_decreasing`, `non_monotonic`, or `n/a` for categorical features), `peak` (`null` for monotonic or categorical features, otherwise an object {`type`: `max`|`min`, `x`, `y`}), `importance_rank` (1–9), and for categorical features `top_categories` / `bottom_categories` (each a list of up to three {`cat`, `y`} objects). Derived numeric fields `effect_range` and `sign_change_x` and an optional free-text `note` are also stored. Example (the actual record on disk for `ebm/hr`): {`form`: `categorical`, `direction`: `categorical`, `monotonicity`: `n/a`, `peak`: `null`, `importance`: 0.889, `importance_rank`: 1, `effect_range`: 4.027, `top_categories`: [{`cat`: "17", `y`: 1.215}, {`cat`: "18", `y`: 1.148}, {`cat`: "8", `y`: 1.109}], `bottom_categories`: [{`cat`: "4", `y`: −2.812}, {`cat`: "3", `y`: −2.183}, {`cat`: "5", `y`: −1.718}]}. The judge sees `form`, `direction`, `monotonicity`, `importance_rank`, `peak`, `sign_change_x` (as a formatted `sign_change` string when present), `top_categories`, `bottom_categories` and `note` (`utils/eval.py` `_global_reference`). The field `effect_range`

is not exposed. `sign_change_x` records only the first zero crossing. `direction` and `monotonicity` are computed before the near-flat override and are not changed by it.

C Rubric Definition

The rubric implementation is `utils/rubric.py`. The three sub-scores are in $[0, 1]$. Direction: a keyword match against a small vocabulary of direction terms, checked against the ground-truth direction. A spurious peak claim on a non-monotonic feature yields partial credit. Rank: extracts a numeric rank via `utils/rubric.py _ extract_rank`, which collects candidate "(position, value)" tuples across five phrasing families, numeric with a rank cue ("rank 3", "ranked 3rd", "#3", "N of 9" and "N out of 9"), word ordinals following "rank(ed)" ("ranked third"), numeric ordinals combined with "most/least" ("the 3rd most important"), word ordinals inside anchoring phrases ("the third-most important", "it is the second most important"), and bare superlatives ("the least important", "by far the strongest"), and returns the earliest by text position, so a comparison to another feature named later in the sentence does not overwrite the subject's own claim (see `tests/test_rubric_rank.py`, 69 real and constructed cases including four verbatim quotes from the actual records). The extracted rank scores 1.0 for an exact match, 0.5 off by one, 0.25 same coarse tier (top-3 / middle / bottom-3), 0.0 otherwise. A description with no parseable rank scores 0.0. Structure: form-specific keyword and pattern matching against the ground-truth form and, where applicable, the top categories.

D Full Per-Feature Results

The raw records live under `results/`: the G2a set in `global/` (72 files), the whole-model set in `global_whole/` and its per-feature split in `global_whole_split/` (10 records, 90 after splitting), the 72 variance draws in `global_variance/`, and the judge outputs in `global_judge*/`. The compact table driving the G2a scoring parts of Sections 5.3–5.7 is `results/global_rubric.csv`. The pipeline summary file `results/eval_summary.csv` belongs to the local instance-level pre-study (Appendix E), not to the global track, and is not the source of the G2a/G2b tables in this paper.

E Local Pre-Study (n = 20)

The instance-level pre-study is retained as design evidence for the shift of unit of analysis (Section 3.2). Three of the four pipelines saturate at the maximum faithfulness score of 5.00, which is why it is not a co-equal result. Table 17 reports the summary metrics. No prose analysis is given. One caveat applies to the EBM side. Its local contributions exclude the 15 interaction terms (Section 4.4), so $\exp\big(\beta_0 + \sum_j f_j(x_j)\big)$, the reconstruction the local prompts describe to the LLM, does not reproduce the EBM's

Table 16. All twelve pairwise Wilcoxon signed-rank tests on judge faithfulness (18 pairs). $n_{\neq}$: pairs with different scores (the others are dropped), $+{:}-$: pairs favoring the first / second format, p approx.: normal approximation (used in the text), p exact: sign-flip permutation over the non-zero pairs. Holm_6: within one judge. Holm_{12}: across both judges. r_{rb}: matched-pairs rank-biserial correlation (paired effect size). δ: Cliff's delta (unpaired, positive values favor the first format).

Judge	Pair	$n_{\neq}$	$+{:}-$	p approx.	p exact	Holm_6	Holm_{12}	r_{rb}	δ
Anthropic	template–json	8	4:4	0.521	0.586	1.000	1.000	+0.25	+0.07
Anthropic	template–vision	6	2:4	0.595	0.625	1.000	1.000	−0.24	−0.10
Anthropic	template–tooluse	9	4:5	0.809	0.914	1.000	1.000	+0.09	−0.03
Anthropic	json–vision	4	0:4	0.063	0.125	0.380	0.570	−1.00	−0.17
Anthropic	json–tooluse	8	3:5	0.471	0.617	1.000	1.000	−0.28	−0.10
Anthropic	vision–tooluse	4	3:1	0.257	0.500	1.000	1.000	+0.60	+0.06
OpenAI	template–json	16	13:3	0.005	0.004	0.030	0.061	+0.78	+0.57
OpenAI	template–vision	13	10:3	0.022	0.024	0.108	0.238	+0.70	+0.47
OpenAI	template–tooluse	13	9:4	0.036	0.034	0.142	0.356	+0.65	+0.33
OpenAI	json–vision	11	5:6	0.319	0.356	0.956	1.000	−0.33	−0.15
OpenAI	json–tooluse	11	4:7	0.365	0.418	0.956	1.000	−0.30	−0.18
OpenAI	vision–tooluse	10	4:6	0.957	0.863	0.957	1.000	+0.02	−0.05

prediction. Across the ten EBM instances the gap on the log scale ranges from -0.11 to 0.47, which means the main effects alone miss the prediction by up to a factor of 1.6. For XGBoost the SHAP values reproduce the prediction exactly.

Table 17. Local pre-study ($n = 20$ instances per pipeline). Judge faithfulness is on a 1–5 scale. Reported as design evidence for the shift of unit of analysis. Three of four pipelines are at the ceiling.

Pipeline	Words	Tokens in	Tokens out	Judge faithfulness	Tool calls
Template	57	0	0	5.00	n/a
JSON to Text	250	600	524	5.00	n/a
Vision	239	1 085	571	4.50	n/a
Tool Use	403	5 786	1 268	5.00	6.2

F Variance Study

The full spread table per cell across the three draws is in `results/global_variance/`. The within-cell SDs and the modality spans in Table 10 are re-derivable from those files with the code in `notebooks/04Gf_Generation_Variance.ipynb`. The two Monte Carlo runs behind Section 5.4 (a 24-cell subset run and a full mean-of-18 run) are implemented in `analyses/monte_carlo_orderings.py` (run with `--variant 24` or `--variant 18`). Both

use `random.seed(42)` and `random.choice` per cell, take the Anthropic G2a judge scores for the ten non-redrawn cells in the full-18 case, and exclude the truncated `json_xgb_weekday` draw. `analyses/numbers_ledger.py` recomputes these quantities from `results/` and merges the Monte Carlo outputs. The 24 cells are {`windspeed`, `holiday`, `weekday`, `hr`} × {EBM, XGB} × {`json`, `vision`, `tooluse`}. `hr` is the ceiling control.

Monte Carlo shares. An iteration counts as tied when two or three per-format means coincide. In the 24-cell run 37% of the 5 000 iterations are tied. Among the untied ones `vision` finishes strictly first in 54% (`tooluse` 39%, `json` 7%), the modal strict ordering `vision` > `tooluse` > `json` holds in 42.4%, and the mean best-to-worst spread per iteration is 0.409. In the full-18 run 22% of iterations are tied. `vision` is strictly first in 59% of all iterations, `tooluse` in 18% and `json` in 0.3%, and `json` is strictly below both other formats in 89% (ties between the other two allowed). Among the 78% untied iterations `vision` is strictly first in 76%, `tooluse` in 24% and `json` in 0.4%. All six strict orderings also occur on this scale, but three of them in only 0.06–0.22% of iterations each, so the "all six occur" statement rests on the 24-cell run.

G Limitations

The conclusion-bounding subset is summarized here.

Design. XAI method is confounded with prediction model (RQ5 descriptive only). The template baseline writes out the ground truth in fixed phrases. Its phrase map describes categorical features as "non-monotonic" and names no lowest categories, which costs it points on both judges, most on the Anthropic one (Section 5.3). The beeswarm image plot renderings differ between the two arms, and any `vision_beeswarm` EBM/XGB difference is partly rendering. Each panel has its own auto-scaled y-axis, so scale is not a cross-feature cue even though each title prints the feature's importance.

Measurement. The rubric and the judge share the ground truth and are not independent. Completeness saturates on G2a. The OpenAI judge is stricter on the LLM formats (0.83 / 0.78 / 0.61 for json / vision / tooluse) and scores the template 0.22 higher than the Anthropic judge does. Part of the judge variance is judge-side rather than generation-side. Beyond vendor, the two judges differ in model generation (`gpt-4o`, released in 2024, is older than `claude-opus-4-8`) and in decoding. The OpenAI judge runs at $\texttt{temperature} = 0$. The Anthropic judge does not accept the `temperature` parameter and runs at the API default (Section 3.5). The vendor split in Section 5.6 is therefore not identified against self-preference, deduction strictness, or these model and decoding differences. On G2a the tool-use records are judged with the retrieval transcript attached (`utils/eval.py _tool_trace_block`, with the note that retrieved values may count as evidence). JSON and vision records are judged from the visible answer alone, so the four G2a handovers are not on symmetric evidence. On G2b the split records carry no `tool_calls` field, so no trace is attached and the RQ4 pull-vs-push comparison is not

affected by this asymmetry. A separate rule mismatch is that the per-feature generation prompt lacks the near-flat/"negligible" rule the judge rubric deducts against (Section 5.3).

Statistics. Single draw per cell. A mixed-effects model with feature as a random effect would be cleaner. No equivalence test is run for the null. On the whole-model track this is sharper still. Each of the ten G2b cells (5 conditions $\times$ 2 XAI arms) is a single whole-model generation, split into 9 records for scoring. Every 18-per-condition row in Tables 11 and 12 and the "90/90" count are slices of 2 draws, not 2×9 independent draws. A 24-cell Monte Carlo on G2a shows all six modality orderings occur (Section 5.4), and the same fragility attaches to every G2b cell.

Scope. Single dataset (bike sharing), single generator (Claude Sonnet 4.6), $\mathrm{temperature} = 1.0$.

The aggregated curves average contributions per unique feature value without a minimum count, so in sparse tails a single observation can set the peak, sign change or effect range (`xgb/windspeed`: the five highest values have one observation each).

The binary features `yr` and `holiday` are treated as continuous curves on two grid points, so their `sign_change_x` (e.g. 0.51 for `ebm/yr`) is a zero crossing interpolated between the levels 0 and 1 with no substantive meaning. It is shown to the judge but adds nothing beyond the sign of the two levels. For `weathersit`, which has four categories, the three highest and the three lowest categories overlap in two entries.

The prompts and the ground-truth labels convert normalized temperature as value $\times$ 41 to °C, whereas the current dataset documentation defines the hourly values as $(t + 8)/47$. Scores are unaffected because the ground truth, the template and the LLMs share the conversion, but temperatures stated in the generated texts are off by 2 to 8 degrees.

The per-feature JSON prompt defines the peak as "the x at which y is largest". For XGBoost the payload is the raw scatter of 12 152 points, so the instructed peak is the single largest SHAP value rather than the aggregated peak the ground truth uses. In the judge prompt the MEDIUM anchor example computes a score of 2 and then keeps 3, and all three anchor examples use `temp`.[2] There is no near-flat anchor, although the near-flat stratum is where the two judges diverge most.

The rubric's direction and structure sub-scores match descriptions against a hand-curated keyword list drafted with Claude assistance. Because the generator comes from the same model family, phrases the rubric recognizes may be more likely to appear in the LLM output than in a template or in a human description. That is a possible circularity on the rubric side.

H AI Usage Declaration

Two kinds of AI use have to be kept apart. First, LLMs are the object of study: Claude Sonnet 4.6 generates the explanations, and `claude-opus-4-8` and `gpt-4o` act as judges (Sections 3.2 and 3.5). This use is part of the method and is documented there.

[2] `prompts/judge_system_global.md`.

Second, the author used Claude (Anthropic), mainly through Claude Code, as an assistant for the following tasks:

- *Programming.* The research design, the pipeline architecture and the experimental conditions were specified by the author, partly together with the supervisor. Most of the pipeline, analysis and test code that implements them was generated by the AI from the author's instructions and refined over many iterations ("vibe coding"). Where the AI proposed alternatives for the implementation or the design, the author decided whether to adopt them. The author directed, ran and tested the code. All numbers reported in this paper are computed by this code from `results/` and checked by the test suite and `analyses/numbers_ledger.py` (Section 4.6). None is taken from chat output.
- *Prompt development.* The generation and judge prompts in `prompts/` (Appendix A) were refined with AI suggestions, and the author decided on the final wording. Because the assistant belongs to the same model family as the generator and the Anthropic judge, this is a further possible path for the self-preference effect discussed in Section 5.6.
- *Rubric keyword lists.* The direction and structure keyword lists of the rubric (`utils/rubric.py`) were drafted with AI assistance and reviewed by the author. The possible lexical bias is discussed in Section 3.5 and Appendix G.
- *Literature search.* In parts, the AI was used to find candidate sources.
- *Manuscript review and revision.* Drafts of this paper were reviewed with AI for grammar, spelling and style, and for the consistency of the argument and of the reported numbers. In revision rounds the AI also drafted short new passages directly in the manuscript, following the author's instructions and review comments. The author checked, edited and approved every such passage, and adopted, changed or rejected every proposed correction.

The author takes full responsibility for the entire content of this paper, including all parts produced with AI assistance.